

A Nested Basket Model for the dunnhumby Complete Journey Data

What the data says, what the model does about it, and what still fails

Abstract

We build a model of grocery baskets that can answer price counterfactuals and generate synthetic shopping data. The document is organised around one discipline: every design choice is justified by a measurement on the data, and every claim about the model is a number from a held-out test set. We first describe what the data actually contains (Section 3), then the model (Section 4), then an explicit map from each data finding to the model component that answers it (Section 5), then experiments (Section 6). Section 7 lists what the model still gets wrong. The headline results: the model beats the strongest simple baseline by +0.342 nats, its price coefficient survives a structural placebo (retaining 0.0%), and synthetic baskets it generates carry 87% of the real price elasticity. The main remaining failure is that generated baskets contain roughly the right *amount* of co-purchase structure but put it on the wrong *pairs*.

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1 The problem

A retailer wants to know: if I cut the price of one product, what happens? Not just to that product, but to the whole basket. Do people buy more of it, switch to it from a competitor, or start buying the category at all when they otherwise would not? And can we answer this for one household, so that coupons can be targeted?

Answering this needs a model of the whole basket. Three quantities matter and they are different from each other:

- **Incidence.** Does this household buy from this category on this trip?
- **Allocation.** Given that it does, which product does it pick?
- **Quantity.** How many units, and how many different products from the category?

A model that only answers “which product” cannot distinguish a price cut that grows a category from one that merely moves share within it. That distinction is the retailer’s actual question.

2 Data

The dunnhumby “Complete Journey” data: two years of transactions from a grocery chain. After filtering, the working dataset is:

quantity	value
households N	2,066
items J	5,455
categories C	188
sub-commodities S	758
stores	115
days D	712
(basket, item) rows	1,566,063
baskets	199,345

The only item filter is at least 100 purchase lines, which is statistical support for that item’s own parameters. Households need 20–300 trips. Nothing is dropped to protect a modelling assumption.

Splits are by calendar week, not at random. Train is week < 83 , validation 83–90, test ≥ 91 . A model whose point is to answer “what happens next week” must be scored on later weeks. A random split would let it see each household’s future.

Prices are deviations, not levels. For item j on day t we use

$$\Delta \log p_{jt} = \log p_{jt} - \frac{1}{D} \sum_{t'=1}^D \log p_{jt'}, \quad (1)$$

so an item's price *level* is absorbed by its own intercept and what identifies the response is that item moving against its own normal price. Where a store-level price was observed we add the store deviation on top; otherwise it falls back to the chain price.

3 What the data says

This section reports measurements taken *before* any model, using nothing but counts and least squares. Each subsection ends with the design consequence.

3.1 Demand responds to price

Build a panel with one row per (item, week). Let b_{jw} be the number of purchase lines and T_w the number of baskets chain-wide that week. Define

$$y_{jw} = \log \frac{b_{jw} + 0.5}{T_w}, \quad x_{jw} = \text{mean } \log p_{jw}. \quad (2)$$

The 0.5 keeps zero-purchase weeks in the panel. The elasticity is an ordinary least squares slope after removing each item's own mean from both sides — a *within-item* regression, so the comparison is an item against itself over time rather than expensive items against cheap ones:

$$\hat{\varepsilon} = \frac{\sum_{j,w} (x_{jw} - \bar{x}_j)(y_{jw} - \bar{y}_j)}{\sum_{j,w} (x_{jw} - \bar{x}_j)^2} \quad \text{over 556,410 item-weeks.} \quad (3)$$

left-hand side	elasticity	what it measures
log buyers per trip	−0.792	whether a household buys the item at all
log units per trip	−0.945	total units moved
log units per buyer	−0.235	how much a buyer takes, given that they buy

A 10% price cut raises total units by about 9.5%. Of that, $-0.235 / -0.945 = 0.25$ comes from existing buyers taking *more*, and the rest from more households buying at all.

Consequence. A model that treats purchase as binary discards a quarter of the price response. Quantity needs its own component.

3.2 A price cut moves three separate margins

The units bought from a category factor exactly:

$$\underbrace{\frac{\text{units}}{\text{basket}}}_{\text{total}} = \underbrace{\frac{\text{visits}}{\text{baskets}}}_{\text{incidence}} \times \underbrace{\frac{\text{lines}}{\text{visit}}}_{\text{breadth}} \times \underbrace{\frac{\text{units}}{\text{line}}}_{\text{depth}}. \quad (4)$$

Taking logs, the three elasticities must add to the total. Estimating each with Eq. 3 on a (category, week) panel of 15,309 cells, against the mean price of *every* item in the category that week:

margin	quantity regressed	elasticity
incidence	log(visits/baskets)	−0.675
breadth	log(lines/visit)	−0.092
depth	log(units/line)	−0.158
sum		−0.925
independent estimate (Eq. 3, item panel)		−0.945

The sum matches the independently measured total to 0.021. Two panels, two units of analysis, no shared code path. About 73% of the entire units response is households deciding to enter a category at all.

Consequence. Incidence is the dominant margin and must be modelled explicitly.

3.3 Promotions: an event study

Find every item-week where the log price fell by at least 0.15 against the previous week. There are 37,132 such events over 4,712 items. Normalise demand by the item’s own mean so a staple and a niche item can be averaged together:

$$\text{norm}_{jw} = b_{jw} / \bar{b}_j. \quad (5)$$

Events are not independent: an item is cut about eight times, and 57.7% of consecutive cuts for the same item fall less than seven weeks apart, so their windows overlap. We therefore also report **isolated** events — no other cut of that item within ± 3 weeks — of which there are 20,725.

Neither 1.0 nor the pre-period is a valid baseline. norm averages exactly 1.0 over all of an item’s weeks *including its own promotions*, and the pre-period is selected to be expensive by the event definition. We use a **quiet week**: an item-week more than three weeks from any cut of that item. There are 368,173, and they run at 0.896 of the item’s mean demand at essentially its normal price.

weeks from cut	demand	price (log dev.)	demand $\div$ quiet
−3	0.925	+0.050	1.03
−2	0.859	+0.098	0.96
−1	0.870	+0.123	0.97
0	2.136	−0.194	2.38
+1	1.337	−0.139	1.49
+2	1.117	−0.055	1.25
+3	1.065	−0.003	1.19
quiet week	0.896	+0.011	1.00

Three readings.

The pre-period does not ramp. Demand before the cut is within 3% of the quiet level. If retailers timed promotions to demand that was already rising, the spike would be partly its own cause. It is not.

The promotion week is far steeper than elasticity alone. From −1 to 0 the price moves −0.317 in logs and demand moves $\log(2.136/0.870) = +0.898$, an implied slope of −2.83 — $3.4\times$ the −0.792 measured on ordinary week-to-week variation. A promoted price arrives with an endcap and a shelf tag that the price variable does not carry.

Most of the tail is the promotion still running. At +1 the price is still 0.139 below normal. By +3 the price has returned and demand is 1.19 of quiet. No offset dips below quiet, so no stockpiling is visible within three weeks.

3.4 Households differ, and the difference is measurable

Taste. Split each household’s purchases in half by date. Count purchases per category in each half, giving two vectors in $\mathbb{R}^{188}$, and scale each to unit length. Compare with a dot product (a cosine, since both are unit length). Averaging over 2,066 households and comparing against *every* cross-household pair:

	cosine
a household’s own two halves	0.7838
two different households, all 4,266,290 pairs	0.4710
ratio	1.66×

A stranger already scores 0.4710 because everyone buys milk and bread; the *gap* is what is personal.

Price sensitivity. For each household, regress $\log(\text{units})$ on $\log(\text{price})$ using only (household, item) pairs bought three or more times, after subtracting each pair’s own means:

$$\hat{\beta}_i = \frac{\sum_{j,t} \tilde{x}_{ijt} \tilde{y}_{ijt}}{\sum_{j,t} \tilde{x}_{ijt}^2}, \quad \tilde{x}_{ijt} = \log p_{jt} - \overline{\log p_{ij}}, \quad \tilde{y}_{ijt} = \log u_{ijt} - \overline{\log u_{ij}}. \quad (6)$$

Across 1,640 households the median is -0.169 with p10–p90 of -0.427 – -0.016 . But the spread alone proves nothing: noisy estimates are spread out even when the truth is identical for everyone. Measuring each household twice on disjoint halves gives a split-half correlation of only $+0.236$ on 1,374 households.

Why so low: the estimator divides by $\sum \tilde{x}^2$, the price movement that household actually saw. 18.6% of qualifying rows come from a (household, item) pair whose price never moved. Splitting households by how much price movement they had:

	spread of slopes (sd)	share positive
little price movement	0.226	11.5%
much price movement	0.129	3.5%

Where prices genuinely moved the estimates are tighter and almost all negative. Much of the apparent heterogeneity is bad measurement.

Consequence. Household parameters are justified, but a targeting policy should weight by how much is known about each household.

3.5 Stores are not interchangeable

15.8% of store-item-weeks sit more than a cent from the chain price. More importantly, the median store ever sold only 63% of the catalogue.

Consequence. Scoring an item a store never stocked as a *rejected* alternative is a specification error — it was not there to reject.

3.6 Base rates the model must reproduce

quantity	value
categories per basket, of 188	6.12 (3.25%)
items per basket, of 5,455	7.86
units per basket	10.64
rows buying more than one unit	22.3%, carrying 42.6% of all units
distinct items per purchased category	1.284

4 The model

Household i , day t , week-of-year w , store s , item j , category $c = \text{cat}(j)$.

4.1 Item utility

Everything is built on one quantity:

$$u_{ijt} = \lambda_j + \theta_i^\top \alpha_j + \rho_j^\top \bar{\alpha}_j(S) - (\gamma_i^\top \beta_j) \Delta \log p_{jst} + \eta_j^\top x_{ijt} + \mu_j^\top \delta_w + \zeta_j^\top \xi_s. \quad (7)$$

term	meaning
λ_j	item popularity
$\theta_i^\top \alpha_j$	household taste $\times$ item attributes
$\rho_j^\top \bar{\alpha}_j(S)$	interaction with the rest of the basket
$-(\gamma_i^\top \beta_j) \Delta \log p_{jst}$	price, with per-household sensitivity
$\eta_j^\top x_{ijt}$	how long since this household last bought this kind of thing
$\mu_j^\top \delta_w$	seasonality
$\zeta_j^\top \xi_s$	store $\times$ item affinity

The price coefficient $\gamma_i^\top \beta_j$ is a low-rank bilinear form, so every household gets its own sensitivity and every item its own loading at $(N + J)K_p$ parameters instead of NJ . It enters with a minus sign, so a *positive* value means demand falls when price rises.

4.2 The memory term

Let τ_{ijt} be the days since household i last bought item j 's *sub-commodity*. Then

$$x_{ijt} = \begin{bmatrix} \mathbb{I}\{\text{no previous purchase}\} \\ e^{-\tau/7} \\ e^{-\tau/g_{\text{sub}(j)}} \\ \log(1 + \tau) / \log 100 \end{bmatrix}, \quad (8)$$

with the last three set to zero when there is no previous purchase. g_{sub} is that sub-commodity's median repurchase gap, so "a long time" means the right number of days for milk and for shampoo.

4.3 The basket interaction

For item j in a basket S of n items,

$$\bar{\alpha}_j(S) = \frac{1}{n-1} \left(\sum_{k \in S} \alpha_k - \alpha_j \right), \quad \bar{\alpha}_j(S) := \mathbf{0} \text{ when } n = 1, \quad (9)$$

so the interaction contributes $\frac{1}{n-1} \sum_{k \neq j} \rho_j^\top \alpha_k$: the average affinity between j and each other item in the basket.

Tied versus untied. With $\rho_j = \alpha_j$ the interaction is $\alpha_j^\top \alpha_k$: symmetric, and large exactly when two items are *alike*. That forces co-purchase structure into α , which is what makes the embedding recover product categories (Section 6.6), but it cannot express complementarity — two items that go together *because* they are different. Untying ρ from α follows SHOPPER [1]: the effect of k on j is $\rho_j^\top \alpha_k$, which need not equal $\rho_k^\top \alpha_j$ and may be negative, so complements and substitutes both become representable. Section 6.8 measures the trade.

The joint this implies (tied case). Write b_j for everything in Eq. 7 not involving other basket items, and define for baskets of size n

$$E(S) = \sum_{j \in S} b_j + \frac{1}{n-1} \sum_{\{j,k\} \subset S} \alpha_j^\top \alpha_k. \quad (10)$$

Swapping item j for item m keeps $|S| = n$, so $1/(n-1)$ is identical on both sides and

$$E(S') - E(S) = (b_m - b_j) + \alpha_m^\top \bar{\alpha}_j(S) - \alpha_j^\top \bar{\alpha}_j(S) = u_m - u_j, \quad (11)$$

which is exactly what the choice softmax scores. So the item head’s conditionals are those of $P(S) \propto e^{E(S)}$ at fixed basket size, and Gibbs sampling — resample one slot at a time from the current draft — converges to it. This identity is verified numerically in Section 6.

4.4 The nest

Category volume and item allocation factor as a Poisson–multinomial:

$$Q_{ict} \sim \text{Poisson}(\exp(a_{ic} + \kappa_c IV_{ict})), \quad (12)$$

$$(q_{ijt})_{j \in c} \mid Q_{ict} \sim \text{Multinomial}(Q_{ict}, \text{softmax}_j u_{ijt}), \quad (13)$$

$$IV_{ict} = \log \sum_{j \in c, \text{ stocked at } s} e^{u_{ijt}}. \quad (14)$$

This factorises into independent per-item counts

$$q_{ijt} \sim \text{Poisson}(\exp(a_{ic} + (\kappa_c - 1)IV_{ict} + u_{ijt})), \quad (15)$$

which shows what κ_c means: at $\kappa = 1$ the inclusive value cancels and category volume is whatever the items sum to; at $\kappa = 0$ category volume is fixed and a price cut only moves share; at $\kappa > 1$ the category expands more than proportionally. $\kappa_c = \text{softplus}(\kappa_c^{\text{raw}})$ is initialised at exactly 1.0 so the data must move it in either direction.

4.5 Four heads

Item. A softmax over the true item plus 20 sampled negatives, with unavailable items masked out:

$$\mathcal{L}_{\text{item}} = -\frac{1}{|R|} \sum_{r \in R} \log \frac{e^{u_{jr}}}{\sum_{m \in C_r} \mathbb{1}_{\{\text{avail}_m\}} e^{u_m}}. \quad (16)$$

Quantity. With its own price coefficient, so the two margins need not share one elasticity:

$$z_{ijt} = q_j^0 - (\gamma_i^{q\top} \beta_j^q) \Delta \log p_{jst} + \eta_j^{q\top} x_{ijt}, \quad \text{units} - 1 \sim \text{Poisson}(e^z). \quad (17)$$

Incidence. A Bernoulli per (trip, category):

$$\text{logit}_{ict} = c_c^0 + c_i^{u\top} c_c^c + c_c^{s\top} x_{ict} + \kappa_c (IV_{ict} - \overline{IV}_c) + \underbrace{\sum_{c' \in B, c' \neq c} W_{cc'}}_{\text{optional, Section 6.8}}. \quad (18)$$

Breadth. Given the category is bought, how many *distinct* items:

$$z_{ic}^b = b_c^0 + b_i^u - b_c^p \overline{\Delta \log p_{ct}}, \quad (\text{distinct items}) - 1 \sim \text{Poisson}(e^{z^b}). \quad (19)$$

This head is necessary rather than decorative. Deriving a count from $P(\text{buy})$ alone assumes $Q \sim \text{Poisson}(\lambda)$ with $1 - e^{-\lambda} = p$, giving $\mathbb{E}[Q \mid Q > 0] = \lambda/p$, which is between 1.01 and 1.12 for realistic incidence rates — against a real 1.284.

4.6 Objective

$$\mathcal{L} = \mathcal{L}_{\text{item}} + w_q \mathcal{L}_{\text{qty}} + w_c \mathcal{L}_{\text{inc}} + w_b \mathcal{L}_{\text{brd}} + \frac{1}{B} \left(\ell_2 \|\Theta_{\text{repr}}\|^2 + \ell_2^p \|\Theta_{\text{price}}\|^2 \right). \quad (20)$$

The two penalties are separate on purpose: shrinking the price block at the representation block's rate biases the elasticity toward zero, and the elasticity is the quantity being measured.

5 From measurement to model

Each row is a finding from Section 3 and the component that answers it.

what the data showed	what the model does
25% of the units response is existing buyers taking more	quantity head with its <i>own</i> price coefficient
73% of the units response is entering a category at all	the nest, with κ_c estimated rather than assumed
1.284 distinct items per purchased category; 22.3% of rows buy more than one unit	breadth head; units kept as counts, never binarised
Taste is $1.66\times$ more self-similar within household than across	per-household θ_i , and per-household price sensitivity γ_i
Split-half reliability of per-household slopes is only +0.236	price sensitivity is low-rank ($\gamma_i^\top \beta_j$), not a free per-pair parameter
Median store carries only 63% of the catalogue	availability mask: unstocked items leave the choice set entirely
15.8% of store-item-weeks differ from the chain price	store-level price deviation added where observed
Promotion weeks are $3.4\times$ steeper than ordinary price variation	<i>not</i> addressed; see Section 7
Demand does not ramp before a cut	supports reading the price coefficient causally; tested by placebo

6 Experiments

All numbers are on the held-out test weeks. The seed spread on item log-likelihood is **0.0032** nats, so any gap larger than about 0.01 is real.

6.1 Are the equations what the code computes?

Fifteen checks compare each printed formula against the fitted model — exactly where the document states an identity, by finite difference where it states a derivative. All pass: the utility term by term; the context formula; the $E(S') - E(S) = u_m - u_j$ identity; the Poisson likelihood; $\kappa = \text{softplus}$; the implied breadth λ/p ; the three elasticity derivatives; and that the sampled inclusive value is unbiased for the sum but biased low for its logarithm by Jensen’s inequality.

6.2 Ablations

model	item log-lik	top-1	cost of removing
full model	−1.9927	0.378	—
seed replicate	−1.9895	0.379	—
no store	−2.1820	0.355	0.189
no interaction	−2.1074	0.347	0.115
no household state	−2.0684	0.360	0.076
prices scrambled	−2.0416	0.362	0.049
availability mask only	−1.9932	0.375	0.001
no breadth	−1.9927	0.378	0.000
no quantity	−1.9925	0.378	0.000
no nest	−1.9877	0.380	−0.005

Three heads cost nothing on item ranking, and the nest is marginally better without. This is the design working, not failing: the item head ranks items, while the nest, quantity and breadth heads answer questions the item head never asks. They share only u_{ijt} . A head that improved item ranking *because* it also modelled incidence would mean the two were entangled. Each head must therefore be scored on its own quantity — the breadth head is worth 0.000 nats here and yet fixes the only generation criterion that was failing.

Stores, split honestly. Of the 0.189 nats the store information is worth, **99.7%** is the availability mask, which merely makes the choice set smaller. Modelling stores was still correct, but it is a correctness fix, not an information gain.

6.3 Against simple alternatives

Every model scored through *one* scorer on identical candidate sets: same positives, same negatives, same availability mask, 51,167 held-out purchases with 20 negatives each. Baseline weights are tuned on validation.

model	log-lik	top-1
nested	−2.0011	0.371
nested, no interaction	−2.1195	0.342
household repeat-purchase	−2.3428	0.285
popularity	−2.7375	0.091
random	−2.7378	0.067
household + co-occurrence	−3.0526	0.182
item–item co-occurrence	−3.0655	0.078

The strongest baseline is not popularity but *household repeat-purchase*. Grocery is repetitive and most of what any model can predict is that people rebuy what they always rebuy. The real margin is +0.342 **nats and 8.5 points of top-1**.

Two results cut against the model. Popularity is indistinguishable from random here, because negatives are drawn popularity-weighted — so “we beat popularity” is a weaker claim than it sounds. And raw item–item co-occurrence scores *below* random: the signal exists but counting cannot exploit it against popularity-matched decoys.

What this number is. It is a *conditional* likelihood: the probability of one item given the rest of the basket. Test baskets are from held-out weeks so nothing leaks across time, but this is a fill-in-the-blank score, not “predict the basket”. The from-scratch test is Section 6.7.

6.4 Is the price response causal?

We refit the entire model on a price panel scrambled before fitting, including the store-level deviations — leaving those real would leak genuine prices back in.

model	price coefficient	quantity price coef.	κ
fitted	+0.794	+0.134	0.663
seed replicate	+0.794	+0.109	0.674
prices scrambled	−0.000	−0.000	0.675

The placebo retains 0.0% of the coefficient and costs 0.049 nats of ranking, while leaving κ and the incidence likelihood untouched. Prices scramble; category structure does not.

6.5 Where a price cut goes

From the softmax and the Poisson–multinomial, a 1% cut in item j ’s price moves demand through three channels:

$$\text{allocation: } \frac{\partial \log \pi_j}{\partial \log p_j} = -(\gamma_i^\top \beta_j)(1 - \pi_j), \quad (21)$$

$$\text{incidence: } \kappa_c \frac{\partial IV}{\partial \log p_j} = -\kappa_c (\gamma_i^\top \beta_j) \pi_j, \quad (22)$$

$$\text{quantity: } \frac{\partial \log \mathbb{E}[\text{units}]}{\partial \log p_j} = -(\gamma_i^{q\top} \beta_j^q) \frac{\lambda}{1 + \lambda}, \quad \lambda = e^z. \quad (23)$$

The first two are exact derivatives of the same softmax, which is why they carry $(1 - \pi_j)$ and π_j and why they combine to $-(\gamma^\top \beta)[1 - \pi_j(1 - \kappa_c)]$. Evaluated on 12,724 held-out purchases:

channel	elasticity	share
allocation (share within category)	−0.991	83%
incidence (category expands)	−0.058	5%
quantity (units per buyer)	−0.139	12%
total (mean)	−1.188	

Mostly it moves share. The quantity margin is 12% — a share a binary-purchase model cannot reach at all, and independently corroborated by the model-free units-per-buyer elasticity of -0.235 in Section 3. Note $\kappa = 0.663$ moved across three incidence samplers before settling, so it is weakly identified and no argument here rests on its exact value.

6.6 Do the embeddings recover product structure?

Sub-commodity labels are held out of the model entirely.

	kNN purity	× chance	AUC
fitted α	0.2976	69.6×	0.8222
no interaction	0.2119	49.5×	0.7267
random control	0.0037	0.9×	0.499
popularity control	0.0038	0.9×	0.510

The interaction term is the only component besides the availability mask that materially moves the embedding. Against the paper’s own model on a common 409-item universe, α reaches $12.1\times$ chance and AUC 0.805, against $1.0\times$ and 0.379.

6.7 Generation: does the distribution match?

Comparing three means is not enough, and comparing $\alpha^\top \alpha$ is close to circular — the model is trained to make co-purchased items have large $\alpha^\top \alpha$, so scoring generated baskets with it grades the exam with the answer key. Every check below is **model-free** (item ids and counts only) and **distributional** (a whole distribution or a per-item vector). Generated trips are matched to real ones: same household, day, week and store, so any gap is the model’s. All 24,768 test baskets.

1. Shape. Total variation distance is $\frac{1}{2} \sum_x |P_{\text{real}}(x) - P_{\text{gen}}(x)|$; zero is identical.

	real (mean)	generated (mean)	TVD
items per basket	8.18	9.11	0.30
categories per basket	6.38	6.85	0.30
units per basket	11.06	12.16	0.29
distinct items / category	1.28	1.33	0.06
units per line	1.35	1.34	0.02

The means agree and the distributions do not. Real baskets are far more skewed. Notably the two quantities the heads model *directly* — breadth and units per line — match closely; the three that emerge from composing them do not.

2. Item marginals. Purchase rate of each of the 5,455 items, real against generated: Spearman +0.406, with 0.1% of items bought in the real data never generated. No mode collapse.

3. Co-occurrence lift. For the commonest items, $\text{lift}(j, k) = P(j, k \text{ together}) / (P(j)P(k))$. This is the model-free version of the interaction term.

	mean lift	Spearman vs real
real	3.48	—
generated	1.61	−0.003

4. Held-out labels. Share of within-basket pairs sharing a sub-commodity: real 0.0646, generated 0.0593.

5. Price response recovered from generated data. Rebuild the item $\times$ week panel *from generated baskets* and run Eq. 3 on it. This is the property a pricing policy would be learned from.

	elasticity	item-weeks
real held-out	−0.6480	49,784
generated	−0.5669	56,180

Generated data carries 87% of the real price response — the best of the five checks, and the one that matters most for the stated purpose.

6.8 The co-occurrence investigation

Generated baskets carried only 43% of the real co-occurrence lift. Three separate causes were found, each by measurement.

Cause 1: the wrong level. The item-level Gibbs sweep resamples a slot *from its own category*, because that is the move $E(S)$ is defined over. Splitting generated pairs by whether the sweep can reach them:

pair type	pairs	generated / real lift	Spearman vs real
same category (reachable)	230	49%	+0.290
cross category (not)	4,047	45%	−0.094

95% of item pairs are cross-category, and which categories entered a basket was decided by C independent Bernoulli draws. So most co-purchase structure was set by independent coin flips and no item-level work could reach it.

Cause 2: dilution. $\bar{\alpha}$ divides by $n - 1$, so a partner worth $\alpha_j^\top \alpha_k = 6$ contributes 6/7 in an eight-item basket. Given a free scalar on the interaction, the model sets it to **5.5** — close to the average $n - 1$.

Cause 3: tied embeddings encode similarity, not complementarity. $\alpha^\top \alpha$ averages 4.34 within a sub-commodity and 0.47 across categories. Given a *tied* category-level context the model fitted its scale to -3.0 : it used the term for basket-size competition, because it could not use it for complementarity.

The escape at category level is arithmetic. With $C = 188$, a full symmetric pairwise matrix W is 17,578 free parameters — cheaper than blocks already in the model, and impossible at item level ($J^2 \approx 30M$). W can place two categories at +2 while their embeddings sit far apart.

variant	item log-lik	inc. NLL	κ	lift	same-sub	price
base	−1.9927	0.1104	0.663	43%	57%	82%
+ learnable scale	−1.9966	0.1116	0.658	44%	80%	80%
+ category context	−1.9970	0.1075	0.828	44%	89%	84%
+ pair matrix, weak L_2	−1.9971	0.0987	0.597	49%	121%	70%
+ pair matrix, $L_2 = 0.3$	−1.9966	0.1051	0.665	44%	86%	81%
both, regularised	−1.9970	0.1012	0.815	46%	92%	87%

Co-occurrence competes with price. Unregularised, the pair matrix is the best model of category co-occurrence and drags the price response down to 70%. The mechanism is visible in κ , which falls to 0.597: price reaches incidence *only* through $\kappa \cdot IV$, so any mechanism explaining co-occurrence more cheaply takes that job and the price signal with it. Penalising the matrix separately — the treatment the price block already gets — restores both.

Stacking escapes the trade-off. The tied category context pushes κ *up* while the pair matrix pushes it *down*; together, regularised, the model holds κ at 0.815 and improves four of the five checks at a cost inside the seed spread.

6.9 SHOPPER’s asymmetric interaction, and why it did not help

Section 6.8 concluded that the per-pair failure was a *geometry* problem: $\alpha^\top \alpha$ is symmetric and large exactly when two items are alike, so complementarity — two items co-purchased *because* they differ — is not representable. SHOPPER [1] is built on exactly the mechanism that diagnosis calls for: ρ untied from α , deliberately asymmetric, and free to go negative for substitutes. We also implemented its sequential likelihood, drawing one random permutation per basket so each row sees only the items before it — one unbiased sample from the permutation sum, and a context that matches sequential generation exactly.

variant	item log-lik	inc. NLL	κ	lift	Spearman	same-sub	price
base	-1.9927	0.1104	0.663	43%	-0.003	57%	82%
both, regularised	-1.9970	0.1012	0.815	46%	-0.003	92%	87%
+ untied ρ	-2.0550	0.1011	0.780	47%	-0.005	62%	85%
+ untied ρ , sequential	-2.0645	0.1045	0.816	47%	+0.010	57%	88%

Neither moved the number they were built for. The per-pair rank correlation goes $-0.003 \rightarrow -0.005 \rightarrow +0.010$; that is zero in all three cases. Both cost item ranking (0.058 and 0.072 nats, $18\times$ and $22\times$ the seed spread) and both lose the same-sub co-purchase gain, which follows directly: with ρ carrying the interaction, α no longer has to encode co-purchase structure at all. The freedom was used, not ignored — $\cos(\rho_j, \alpha_j) = +0.31$, so ρ found a substantially different direction. It simply did not find real complements.

This refutes the Section 6.8 diagnosis. The geometry was never the binding constraint: the mechanism that explanation demanded is representable here, was fitted, and bought nothing.

The remaining hypothesis concerns the *training signal* rather than the model. Negatives are drawn unigram^{0.75} from the whole catalogue, so a decoy is almost always separable by $\lambda_j + \theta_i^\top \alpha_j$ before the interaction term is consulted. The softmax is asked “is this item plausible at all”, never “does this item fit *this* basket”. No parameterisation can learn pairwise structure from a loss that does not require it. Drawing a fraction of negatives from the true item’s own category, where popularity and broad taste cannot separate them, is the direct test.

7 What still fails

1. **The pattern of co-occurrence is not learned.** Nine variants — spanning symmetric and asymmetric interactions, tied and untied embeddings, mean and sum pooling, a learnable scale, a category-level context, a free pairwise matrix over categories, and SHOPPER’s sequential likelihood — all leave the per-pair rank correlation at zero (Section 6.9). The model knows baskets should be coherent; it does not know that hot dogs go with buns specifically. The evidence now points at the negative-sampling distribution rather than at the interaction’s functional form.
2. **Promotion weeks are steeper than the model’s price term.** Section 3 measures an implied slope of -2.83 in the cut week against -0.792 on ordinary variation. A promoted price arrives with display and feature that the price variable does not carry, so a single coefficient must average two regimes.
3. **Basket-size distribution.** Means match to a hundredth; the distributions sit 0.30 apart in total variation. Real baskets are far more skewed.
4. **No held-out joint likelihood.** $P(S)$ needs a partition function over all size- n baskets, which is intractable to evaluate. The reported item log-likelihood is conditional on the rest of the basket and is not a substitute.
5. **κ is weakly identified.** It moved with the incidence sampler rather than with the data before settling.
6. **Per-household price sensitivity is barely measurable for light shoppers.** Split-half reliability is $+0.236$ overall and rises with data volume, so targeting should weight by how much is known about each household.

References

- [1] F. J. R. Ruiz, S. Athey and D. M. Blei. *SHOPPER: A Probabilistic Model of Consumer Choice with Substitutes and Complements*. Annals of Applied Statistics, 2020. arXiv:1711.03560.
- [2] S. Donnelly, F. J. R. Ruiz, D. M. Blei and S. Athey. *Counterfactual Inference for Consumer Choice Across Many Product Categories*. Quantitative Marketing and Economics, 2021.